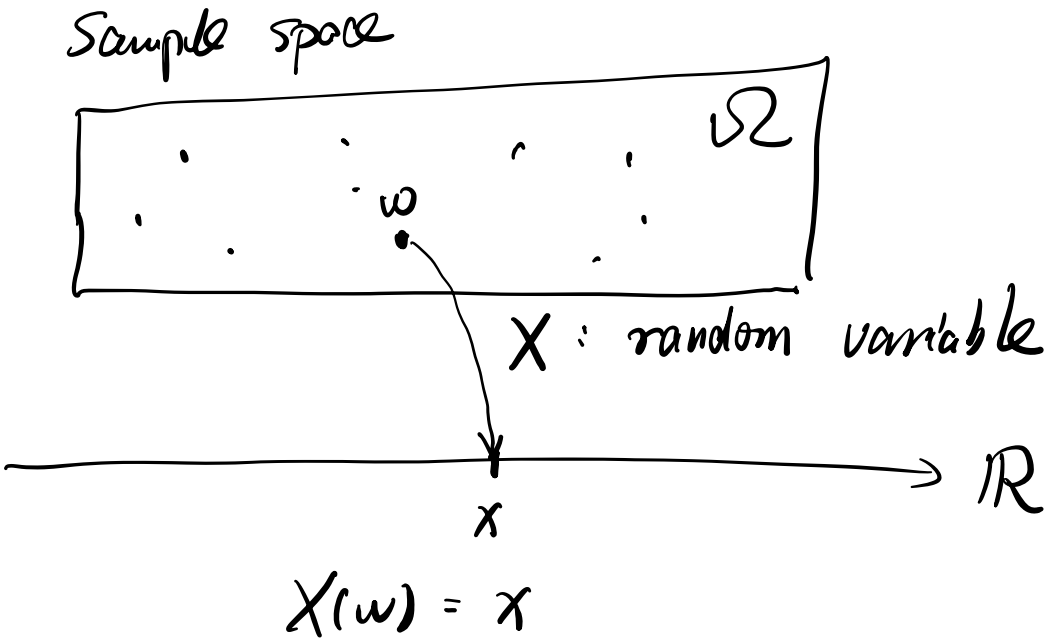
Last Time



If $\{X(\omega) \mid \omega \in \Omega\}$ is countable, we call X a discrete random variable.

$$\{X = x\} = \{\omega \in \Omega \mid X(\omega) = x\}$$

$$P_X(x) = P(X = x)$$

PMF

Families of Discrete Random Variables

So far, we have constructed the PMF of a random variable directly from a particular random experiment. However, many different experiments lead to PMFs with the same mathematical form. Rather than deriving these PMFs from scratch every time, we identify and study several common families of discrete random variables.

Poisson (λ) Random Variable

A discrete random variable X is called a **Poisson random variable** with parameter $\lambda > 0$ if its probability mass function is

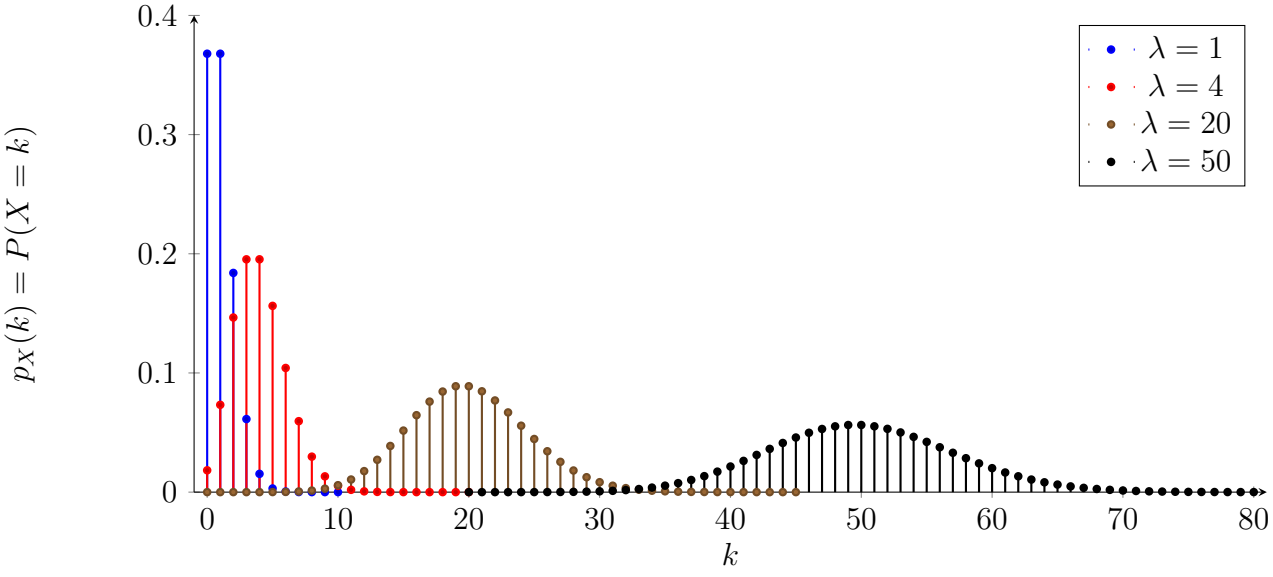
$$p_X(k) = P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

We write

$$X \sim \text{Poisson}(\lambda).$$

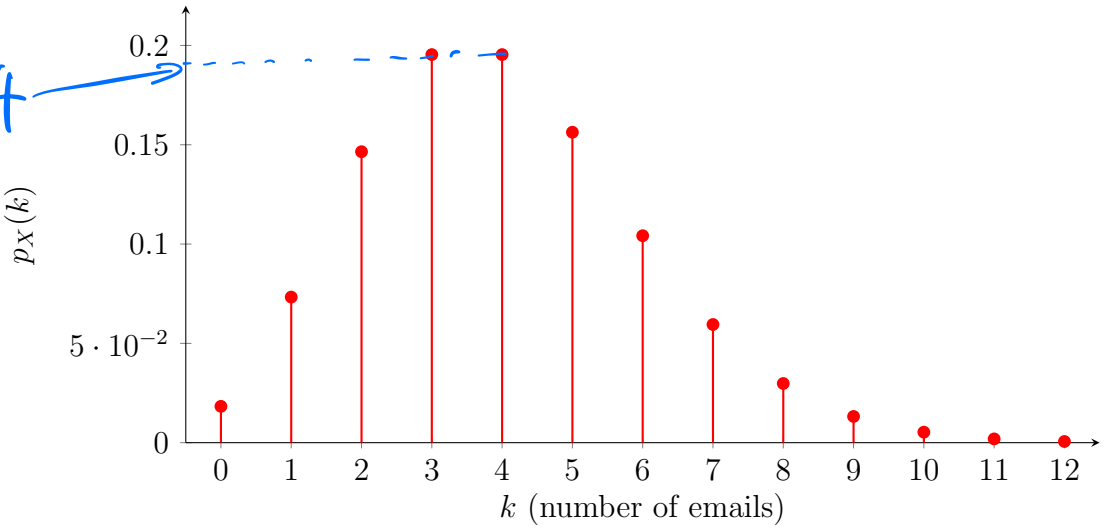
Interpretation of λ.
A Poisson random variable is commonly used to model the number of occurrences of an event in a fixed interval or region when occurrences happen randomly at a known average rate. In this setting,

$$\lambda = \text{average number of occurrences in that interval or region.}$$



PMFs of Poisson random variables for different values of λ.

Notice that λ does more than appear in the formula. It tells us roughly where the probability mass is centered.



0.1954

Probability mass function of $X \sim \text{Poisson}(4)$.

Example. Suppose the number of emails received during a one-hour period is modeled by a Poisson random variable with an average of 4 emails per hour. Let

$\lambda = 4$

X = the number of emails received in one hour.

Then

$$X \sim \text{Poisson}(4),$$

and hence

$$p_X(k) = P(X = k) = \frac{e^{-4}4^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

probability of the event $\{X = k\}$, that is, receiving exactly k emails in one hour.

Compare the probabilities of receiving exactly 3 and exactly 4 emails in one hour.

Sample space $\Omega = \{0, 1, 2, 3, \cdots\cdots\}$

$X : \Omega \to \mathbb{R}$

$\omega \mapsto X(\omega)$

here, $X(\omega) = \omega$

$$p_X(3) = P(X = 3) = \frac{e^{-4}4^3}{3!}$$

$$p_X(4) = P(X = 4) = \frac{e^{-4}4^4}{4!} = \frac{e^{-4} \cdot 4^3}{3!} \approx 0.1954$$

The following example motivates us to introduce another family of random variables: binomial random variables.

Example. Consider an experiment consisting of n independent trials. Each trial results in one of two possible outcomes, called *success* and *failure*. Suppose that the probability of success is p on every trial. Define the random variable

$$X = \text{number of successes in the } n \text{ trials.}$$

What is the probability that $X = k$?

e.g., $(S, S, \cdots S, F, \cdots F)$ (k successes, $n-k$ failures)

An outcome can be represented by a sequence $w = (w_1, w_2, \cdots, w_n)$ where each $w_i \in \{S, F\}$ indicates whether trial i is in a success (S) or failure (F). Hence, the sample space is

$$\Omega = \{S, F\}^n = \underbrace{\{S,F\} \times \{S,F\}}_{n \text{ times}} \times \cdots \times \{S,F\}$$

$$= \{(w_1, w_2, \cdots, w_n) \mid w_i \in \{S, F\}, i = 1, \cdots, n\}$$

For a particular outcome, e.g., $\{\underbrace{S, S, \cdots, S}_{k}, \underbrace{F, F, \cdots F}_{n-k}\}$, the probability of this outcome is

$$p^k (1-p)^{n-k}$$

There are $\binom{n}{k}$ different ways of the n trials result in success.

$$\therefore \quad P(X=k) = \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

Random variables with this PMF occur so frequently that we give this family a name.

Binomial (n, p) Random Variable

A discrete random variable X is called a **binomial random variable** with parameters n and p if its probability mass function is

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

We write

$$X \sim \text{Binomial}(n, p).$$

The parameters have the following interpretations:

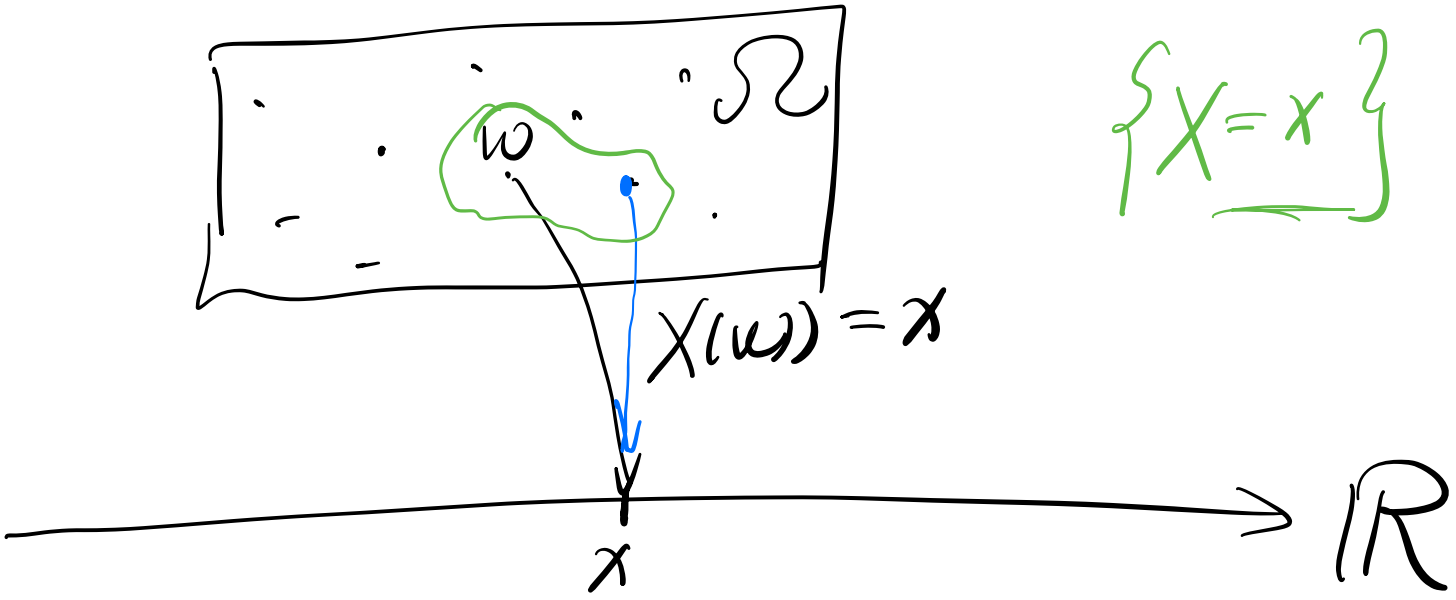
n = number of trials,	p = probability of success on each trial.

independent

Some Common Discrete Random Variables

Distribution	**Notation**	**PMF and Interpretation**
Bernoulli	$X \sim \text{Bernoulli}(p)$	$P(X = k) = p^k(1-p)^{1-k}, \quad k = 0, 1.$ One success/failure trial.
Binomial	$X \sim \text{Binomial}(n, p)$	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \ldots, n.$ Number of successes in n independent trials.
Geometric	$X \sim \text{Geometric}(p)$	$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \ldots.$ Number of trials until the first success.
Negative Binomial	$X \sim \text{NB}(r, p)$	$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, \ldots.$ Number of trials until the rth success.
Hypergeometric	$X \sim \text{Hypergeometric}(N, K, n)$	$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$ Number of successes in n selections without replacement from N objects, of which K are successes.
Poisson	$X \sim \text{Poisson}(\lambda)$	$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots.$ Number of occurrences in a fixed interval or region.

Remark. This table is provided for reference. We will not study all of these distributions in detail. The important idea is that many random experiments lead to probability mass functions belonging to commonly used families of distributions.



CDF $F_X(x)$ cumulates the probability mass distributed by X over the values that are not bigger than x.

Cumulative Distribution Function

Cumulative Distribution Function

The **cumulative distribution function** (CDF) of random variable X is

$$F_X(x) = P(X \leq x)$$

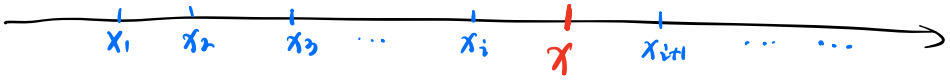
where $x \in \mathbb{R}$ and $P(X \leq x)$ is the probability of the event $\{X \leq x\} = \{\omega \in \Omega \mid X(\omega) \leq x\}$.

Note $\{X=a\}$ and $\{X=b\}$ $(a \neq b)$ are mutually exclusive events.

If X is a discrete RV with range $R_X = \{x_1, x_2, x_3, \cdots\}$

we have

$$\{X \leq x\} = \{\omega \in \Omega \mid X(\omega) \leq x\} = \bigcup_{\forall x_i \leq x} \{X = x_i\}$$



$$\therefore F_X(x) = P(X \leq x) = P(X=x_1) + P(X=x_2) + \cdots + P(X=x_i)$$

$$= \sum_{\forall x_i \leq x} P(X=x_i)$$

Properties of CDF

For any discrete random variable X with range $R_X = \{x_1, x_2, x_3, \cdots\}$ satisfying $x_1 < x_2 < \cdots$, we have

(a) $F_X(x)$ is a non-decreasing function of x.

(b) $\lim_{x\to\infty} F_X(x) = 1$ and $\lim_{x\to-\infty} F_X(x) = 0$.

(c) $F_X(x) = F_X(x_i)$ for all x such that $x_i \leq x < x_{i+1}$.

(d) $F_X(x_1) = p_X(x_1)$ and $F_X(x_{i+1}) - F_X(x_i) = p_X(x_{i+1})$ for $i = 1, 2, \cdots$.

$P(X=x_1)$

Next Time

Recall the following example from *Lecture 07*.

Example. Suppose a fair coin is tossed three times. The sample space is

$$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}.$$

Define the random variable

$$X : \Omega \to \mathbb{R}$$

by letting $X(\omega)$ be the number of heads in the outcome ω. Thus,

$$\begin{aligned} &X(hhh) = 3, \quad X(hht) = 2, \quad X(hth) = 2, \quad X(thh) = 2, \\ &X(htt) = 1, \quad X(tht) = 1, \quad X(tth) = 1, \quad X(ttt) = 0. \end{aligned}$$

The range of X is

$$\{X(\omega) \mid \omega \in \Omega\} = \{0, 1, 2, 3\}.$$

Since the range is finite, X is a **discrete random variable**.

We derived the probability mass function (PMF) of X:

$$p_X(x) = \begin{cases} \frac{1}{8}, & x = 0, \\ \frac{3}{8}, & x = 1, \\ \frac{3}{8}, & x = 2, \\ \frac{1}{8}, & x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Find and sketch the CDF of random variable X.